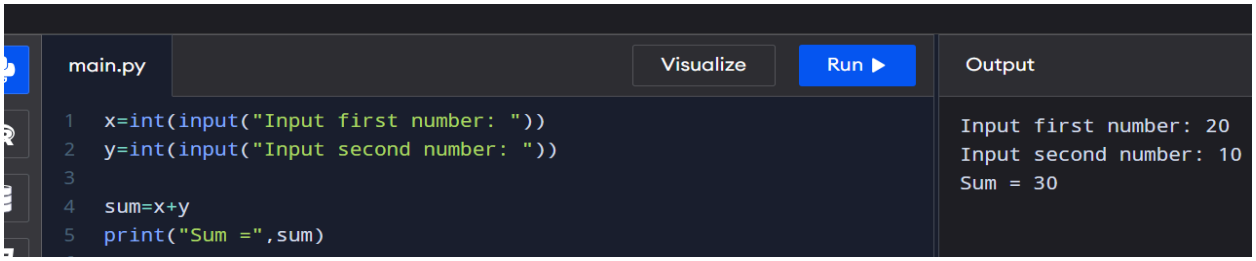
Basic programming exercises and solutions in Python

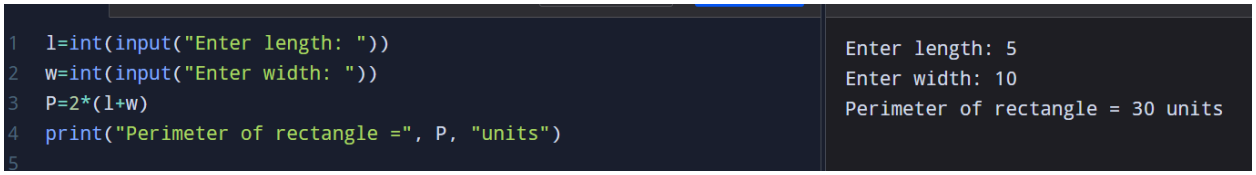
1.



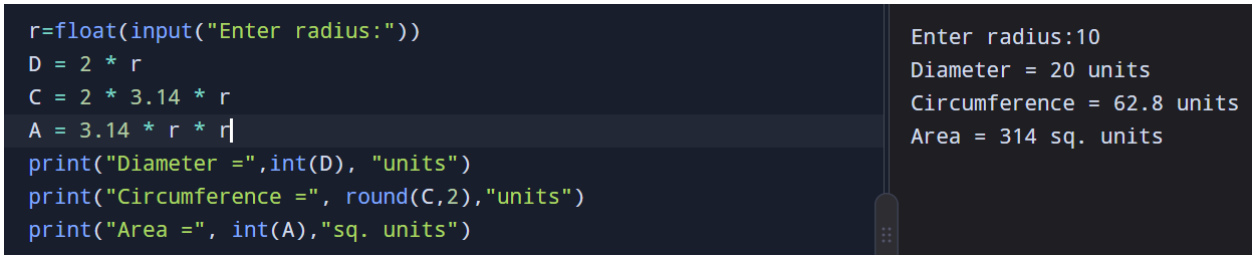
2.



3.



4.



5.

main.py Visualize Run ▶ Output

```
a=float(input("Enter length:"))
b=float(input("Enter width:"))

A= a * b
print("Area of rectangle=", A,"sq.unit")

```

```
Enter length:5
Enter width:10
Area of rectangle= 50.0 sq.unit
```

6.

```
cm=float(input("Enter length in centimeter :"))
m = cm/100
km = cm/100000
print("Length in meter =",m ,"m" )
print("Length in kilometer=", km, "km")
```

```
Enter length in centimeter :1000
Length in meter = 10.0 m
Length in kilometer= 0.01 km
```

7.

main.py Visualize Run ▶ Output

```
c=float(input("Enter temperature in Celsius :"))
f=(c*9/5)+32
print("Temperature in Fahrenheit =",f )

```

```
Enter temperature in Celsius :100
Temperature in Fahrenheit = 212.0
```

8.

```
f=float(input("Temperature in fahrenheit:"))
c=(f-32)*5/9
print("Temperature in celsius =",c,"C")
```

```
Temperature in fahrenheit:205
Temperature in celsius = 96.11111111111111 C
```

9.

```
x=int(input("Enter base:"))
y=int(input("Enter exponent:"))
result = pow (x,y)
print(x, "^", y ,"=", result)

```

```
Enter base:5
Enter exponent:2
5 ^ 2 = 25
```

10.

```
x=float(input("Enter first angle:"))
y=float(input("Enter second angle:"))
z=180-(x+y)
print("Third angle =",z)

```

```
Enter first angle:60
Enter second angle:80
Third angle = 40.0
```

11.

```
x=float(input("Enter base of the triangle:"))
y=float(input("Enter height of the triangle:"))
Area = (x * y) /2
print("Area of the triangle=",Area, "sq.units")

```

```
Enter base of the triangle:10
Enter height of the triangle:15
Area of the triangle= 75.0 sq.units
```

12.

```
a=float(input("Enter side of the equilateral triangle:"))
Area = 1.732 * a * a /4
print("Area of equilateral triangle=", Area,"sq. units")
```

```
Enter side of the equilateral triangle:10
Area of equilateral triangle= 43.3 sq. units
```

13.

```
a=float(input("Enter marks of First subjects :"))
b=float(input("Enter marks of Second subjects :"))
c=float(input("Enter marks of Third subjects :"))
d=float(input("Enter marks of Fourth subjects :"))
e=float(input("Enter marks of fifth subjects :"))

total = a+b+c+d+e
average = total/5
Percentage = total/500*100

print("Total=",total)
print("Average=",average)
print("Percentage=", Percentage, "%")
```

```
Enter marks of First subjects :95
Enter marks of Second subjects :76
Enter marks of Third subjects :85
Enter marks of Fourth subjects :90
Enter marks of fifth subjects :89
Total= 435.0
Average= 87.0
Percentage= 87.0 %
```

14.

```
P=float(input("Enter principle :"))
T=float(input("Enter time:"))
R=float(input("Enter rate:"))
SI=(P*T*R)/100
print("Simple Interest=",SI)
```

```
Enter principle :1200
Enter time:2
Enter rate:5.4
Simple Interest= 129.6
```

15.

```
P=float(input("Enter principle (amount): "))
T=float(input("Enter time:"))
R=float(input("Enter rate:"))
CI=P*(1+R/100)** T
print("Compound Interest=",CI)
```

```
Enter principle (amount): 1200
Enter time:2
Enter rate:5.4
Compound Interest= 1333.0992
```

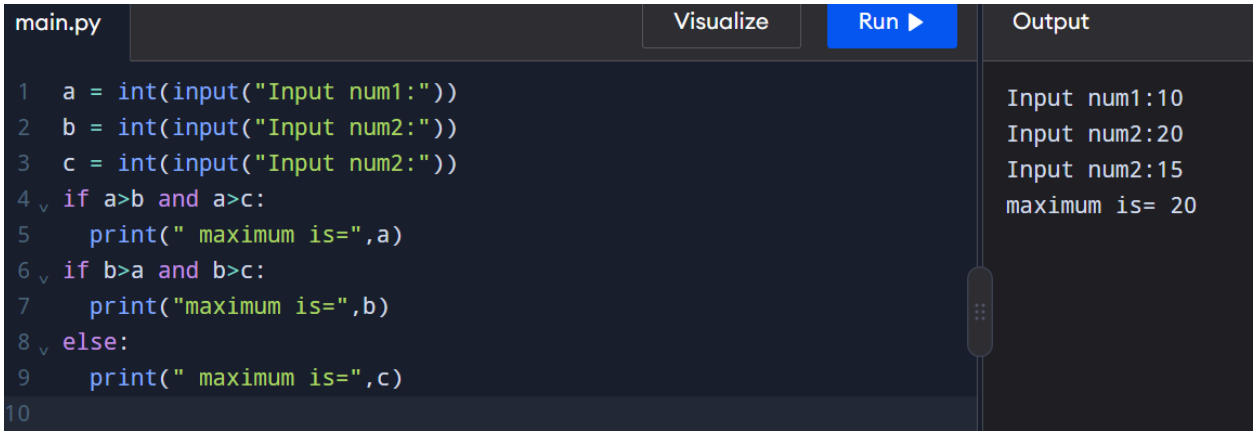
If else programming exercises and solutions in Python.

1.

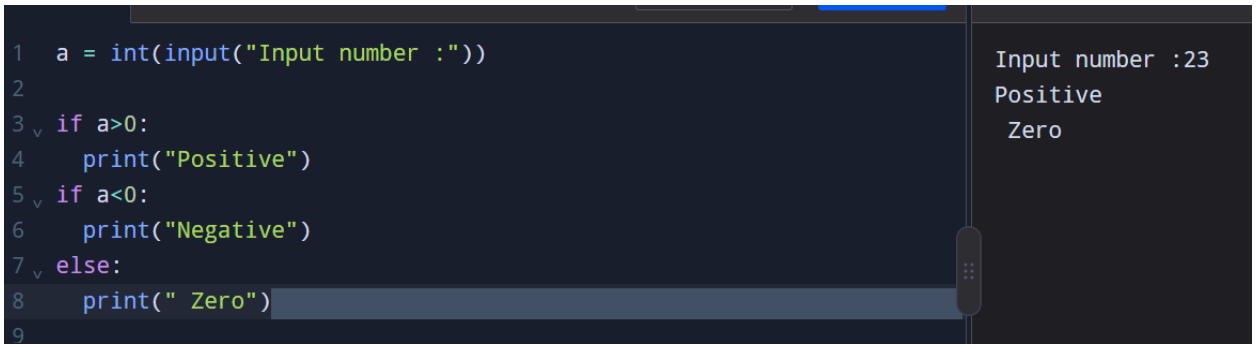
```
a = int(input("Input num1:"))
b = int(input("Input num2:"))
if a>b:
  print(" maximum=",a)
else:
  print(" maximum=",b)
```

```
Input num1:10
Input num2:20
 maximum= 20
```

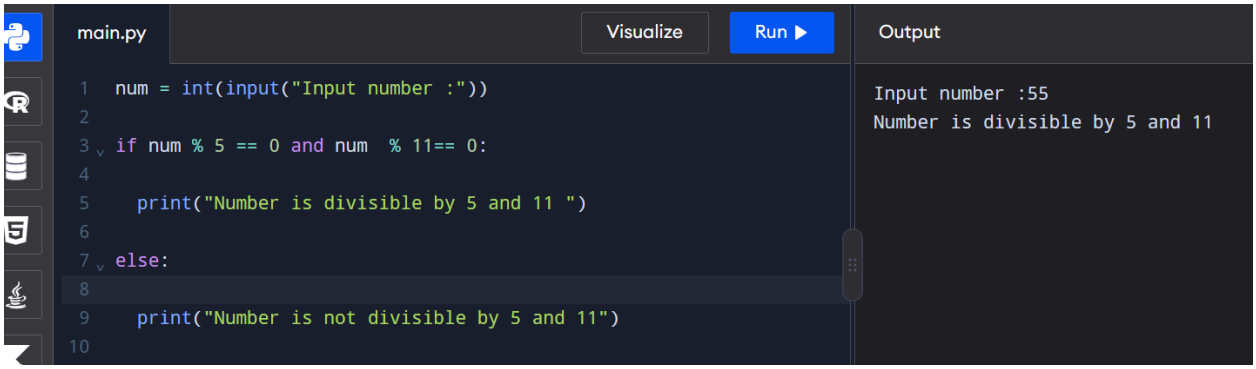
2.


main.py
Visualize
Run
Output
a = int(input("Input num1:"))
b = int(input("Input num2:"))
c = int(input("Input num2:"))
if a>b and a>c:
print(" maximum is=",a)
if b>a and b>c:
print("maximum is=",b)
else:
print(" maximum is=",c)
Input num1:10
Input num2:20
Input num2:15
maximum is= 20

3.


a = int(input("Input number :"))
if a>0:
print("Positive")
if a<0:
print("Negative")
else:
print(" Zero")
Input number :23
Positive
Zero

4.


main.py
Visualize
Run
Output
num = int(input("Input number :"))
if num % 5 == 0 and num % 11== 0:
print("Number is divisible by 5 and 11 ")
else:
print("Number is not divisible by 5 and 11")
Input number :55
Number is divisible by 5 and 11

5.


num = int(input("Input number :"))
if num % 2 == 0:
print("Even number=",num)
else:
print("Odd number=",num)
Input number :10
Even number= 10

6.

```
ch = str(input("Input character :"))

if ch in "aeiouAEIOU":

  print(" 'a' is vowel")

else:

  print(" 'a' is Consonant")

```

```
Input character :a
 'a' is vowel
```

7.

```
ch = str(input("Input character :"))

if ch in "alphabet":

  print(" 'a' is alphabet")

else:

  print(" 'a' is Consonant")

```

```
Input character :a
 'a' is alphabet
```

8.

main.py

Visualize

Run ▶

Output

main.py

```
upper = (input("Input character:"))
if upper.isupper():
  print( " 'C' is Uppercase alphabet")
else:
  print( " 'C'is  Lowercase alphabet")
```

```
Input character:C
 'C' is Uppercase alphabet
```

9.

```
week=int(input("Input week number: "))
if week == 1:
  print("Sunday")
elif week == 2:
  print("Monday")
elif week == 3:
  print("Tuesday")
elif week == 4:
  print("Wednesday")
elif week == 5:
  print("Thursday")
elif week == 6:
  print("Friday")
elif week == 7:
  print("Saturday")
else:
```

```
Input week number: 4
Wednesday
```

10.

```
if month == 1:
  print("31 days")
elif month == 2:
  print("28 days")
elif month == 3:
  print("31 days")
elif month == 4:
  print("30 days")
elif month == 5:
  print("31 days")
elif month == 6:
  print("30 days")
elif month == 7:
  print("31 days")
else:
  print("Invalid month number")
```

```
Inter month number: 5
31 days
```

11.

main.py | Visualize | Run ▶ | Output

```
year = int(input("Input year:"))

if year % 4 == 0:
  print("2004 is leap year")
else:
  print("2004 is  not leap year")
```

```
Input year:2004
2004 is leap year
```

12.

```python
a = int(input("Input first angle :"))
b = int(input("Input second angle :"))
c = int(input("Input third angle :"))

sum = a + b + c
if sum == 180:
  print("The triangle is valid")
else:
  print("The triangle is not valid")
```

```
Input first angle :60
Input second angle :30
Input third angle :90
The triangle is valid
```

13.

```python
a = int(input("Input first side :"))
b = int(input("Input second side: :"))
c = int(input("Input third side :"))

if (a+b) > c:
    if ( b + c) > a:
      if ( a + c) > b:
        print("Triangle is valid")
      else:
        print(" Triangle is not valid")
```

```
Input first side :7
Input second side: :10
Input third side :5
Triangle is valid
```

14.

```python
a = int(input("Input first side :"))
b = int(input("Input second side: :"))
c = int(input("Input third side :"))

if a == b and b == c:
    print("Triangle is equilateral triangle")
else:
    print(" Triangle is not equilateral triangle")
```

```
Input first side :30
Input second side: :30
Input third side :30
Triangle is equilateral triangle
```

15.

```python
sp= int(input("Input selling price :"))

if sp > cp:
  print("Profit =", sp-cp)
else :
  if cp > sp:
    print("Loss =",cp-sp)
  else:
    print("No Profit , No Loss")
```

```
Input cost price :1000
Input selling price :1500
Profit = 500
```

16.

main.py

Visualize Run ▶

```
co = int(input("Input Computer:"))
total = (ph + ch + bi + ma + co)
per= total / 5
if per >= 90:
  print("Grade A")
else:
    if per >= 80:
      print("Grade B")
    else:
      if per >= 70:
        print("Grade C")
      else:
        if per >= 60:
          print("Grade D")
        else:
          if per >= 40:
            print("Grade E")
          else:
            print("Grade F")
```

Output

```
Input Physics:95
Input Chemistry :95
Input Biology:97
Input Mathematics :98
Input Computer:90
Grade A
```